

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Analog and Digital Communications	Course Code:	EE3003
Program:	BS (Electrical Engineering)	Semester:	Fall 2024
Exam:	Assignment 3	Page(s):	3
Chapter(s):	4 (Amplitude Modulation)	Section:	ALL
Submission Date:		CLO:	03
Instructor(s):	Dr. S. M. Sajid, Mohsin Yousuf	Marks:	100

Instructions: The design parts should include the diagram along with mathematical equations.

Question # 1 [CLO 03]

[5+5]

DSB-SC Modulation and Demodulation

(a) An information signal is of the form

$$m(t) = \frac{\sin(2\pi t)}{t}$$

This signal amplitude modulates a carrier of frequency 10 Hz. **Sketch** the DSB-SC waveform and its spectra.

(b) **Analyze** the switching demodulator that uses the ring modulator as a switch with diagrams and mathematical equations.

Question # 2 [CLO 03]

[3+4+4]

Dual-Tone Modulated AM Signal

Consider an AM signal with multi-tone modulation where the modulation signal $m(t)$ is given by

$$m(t) = \cos \omega_m t + 2 \sin 3\omega_m t$$

and the carrier $c(t)$ is given by

$$c(t) = 3 \cos \omega_c t; \text{ where } \omega_c \gg \omega_m$$

Find an expression for (i) $\varphi_{AM}(t)$, (ii) the spectrum of the AM modulated signal, $\Phi_{AM}(\omega)$.

In addition, **sketch** the spectrum of the dual tone modulated signal.

Question # 3 [CLO 03]

[3+2+5]

Modulation with Multiplexing

Two signals $m_1(t)$ and $m_2(t)$, both band-limited to 5000 rad/s, are to be transmitted simultaneously over a channel by the multiplexing scheme shown in Figure 1. The signal at point 'b' is the multiplexed signal, which now modulates a carrier of frequency 30,000 rad/s. The modulated signal at point 'c' is transmitted over a channel.

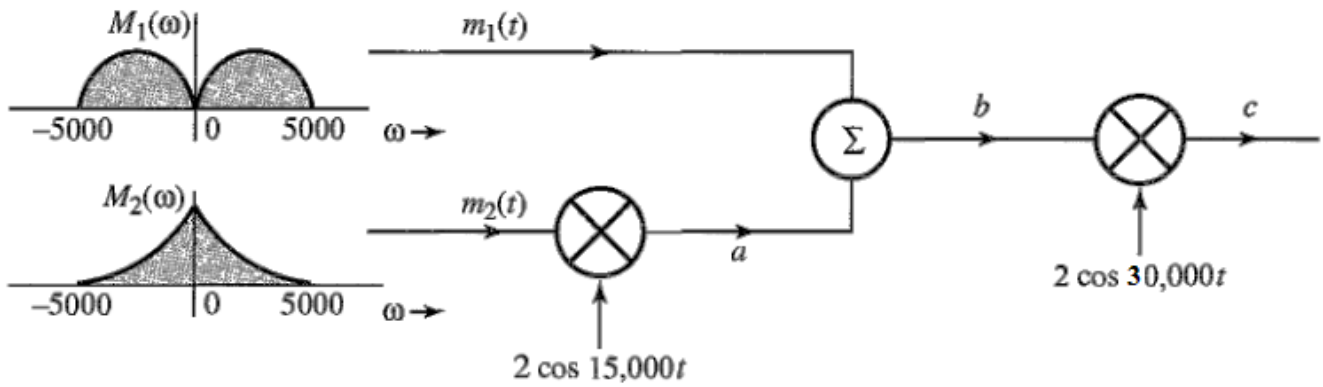


Figure 1. Multiplexing of two signals

- Sketch signal spectra at points a , b and c .
- Find the bandwidth of the channel for transmitting the multiplexed and modulated signal.
- Design a receiver to recover signals $m_1(t)$ and $m_2(t)$ from the modulated signal at point c .

Question # 4 [CLO 03]

[5+5]

Design of DSB-SC modulator

Design a DSB-SC modulator to generate a modulated signal $k m(t) \cos \omega_c t$, where $m(t)$ is a signal band-limited to B Hz. Figure 2 shows a DSB-SC modulator available in the stock room. The carrier generator available generates $\cos^5 \omega_c t$ instead of $\cos \omega_c t$. Would it be possible to generate the desired signal using only this equipment. The filter can be utilized as deemed appropriate.

- Find an expression in time-domain at point b and the *filter* to obtain the required signal at point c .
- Determine the signal spectra at points b , and c indicating frequency bands occupied by these spectra.

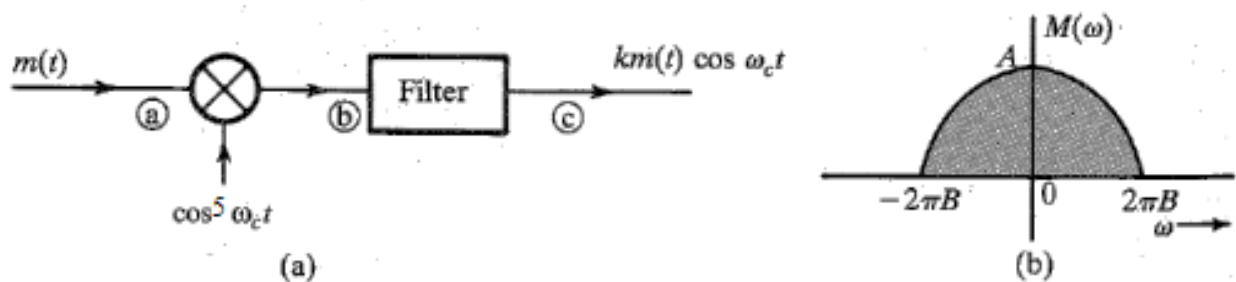


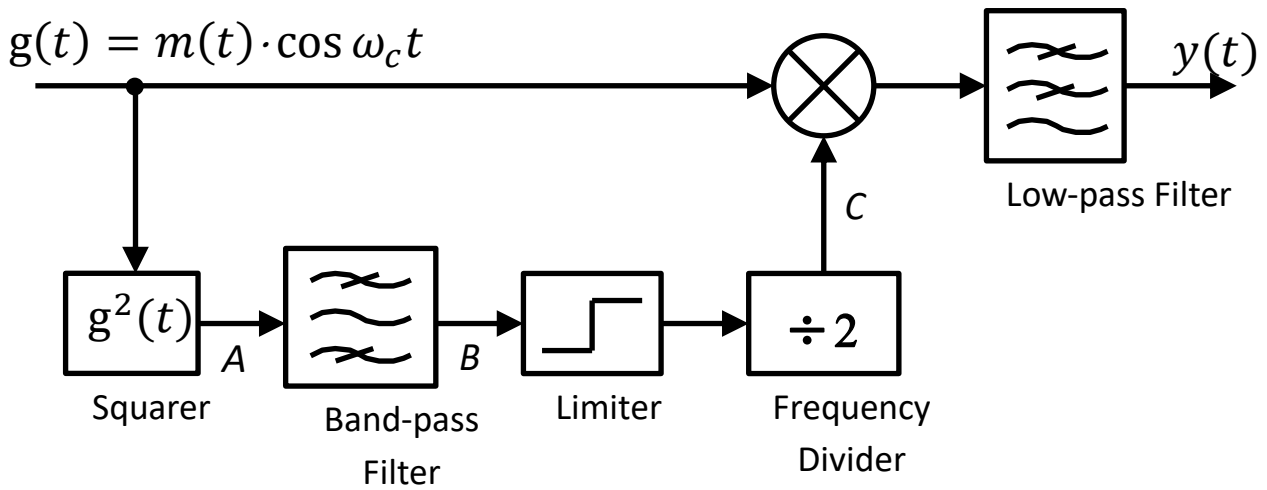
Figure 2.(a) DSB-SC modulator (b) Spectra of $m(t)$

Question # 5 [CLO 03]

[2+2+2+2+2]

“Squaring Loop” Demodulation of DSB-SC

We have an incoming DSB-SC amplitude modulated signal of the form, $m(t) \cdot \cos \omega_c t$. A “squaring loop” is shown in the block diagram.



- If the input is $g(t) = m(t) \cdot \cos \omega_c t$, then **find** the output of the squarer block (that is, output at point "A" in the block diagram)?
- Find** the output of the band-pass filter (that is, at point "B" in the block diagram)?
- Demonstrate** the properties required of the band-pass filter.
- Find** the output $y(t)$.
- Demonstrate** the use of limiter in the block diagram.

Question # 6 [CLO 03]

[5+5]

AM Power and Efficiency

Sketch the conventional AM signal $[A + m(t)] \cos \omega_c t$ for the periodic triangle signal $m(t)$ shown in Figure 3 corresponding to modulation index: $\mu = 0.8$, $\mu = 1$ and $\mu = 2$.

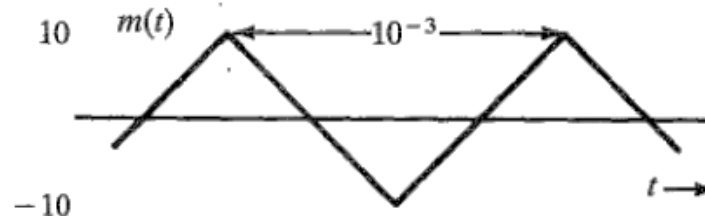


Figure 3. A periodic triangular message signal $m(t)$

For $\mu = 0.8$ case,

- Find** the amplitude and power of the carrier.
- Find** the sideband power and the power efficiency η .

Question # 7 [CLO 03]

[5+5]

An amplitude modulated signal is given by

$$\varphi_{AM}(t) = 2.5 [b + 3 \cos \omega_m t] \cos \omega_c t, \quad \text{for } \omega_m \ll \omega_c$$

Find the power efficiency η of the AM signal for, (i) $b = 8$ and (ii) $b = 3$.

Single Sideband (SSB)

Find $\phi_{LSB}(t)$ and $\phi_{USB}(t)$ for the modulating signal $m(t) = B \text{sinc}(2\pi Bt)$ with $B = 1000$ and carrier frequency $\omega_c = 10,000\pi$. Follow these steps.

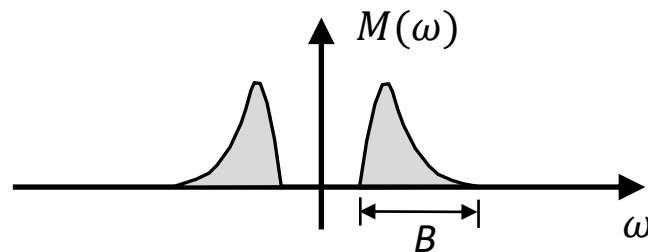
- Sketch spectra of $m(t)$ and corresponding DSB-SC signal $2m(t) \cos \omega_c t$.
- Find and sketch the LSB spectrum.
- Find and sketch the USB spectrum.
- Find the bandwidth of the SSB signal.

Question # 9 [CLO 03]

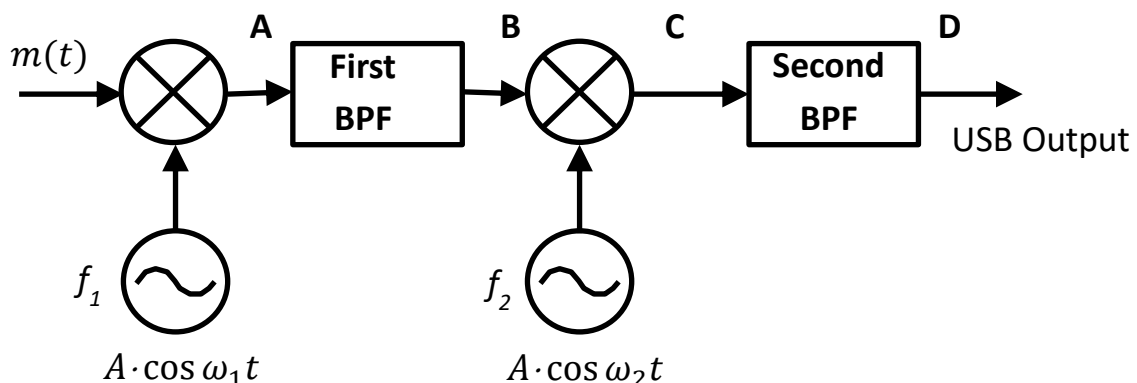
[2+3+3+2]

Two-Step SSB Generation

Another method to generate SSB signals is the Two-Step SSB Generator. This approach can be used when the modulating function $m(t)$ has a two-sided frequency spectrum $M(\omega)$ with negligible energy around zero frequency (i.e., around DC) as shown in the spectral plot $M(\omega)$ below. Note the spectrum $M(\omega)$ has a gap centered at $\omega = 0$ (meaning DC).



The two-step SSB modulator (or generator) produces an USB signal with carrier frequency $f_c = f_1 + f_2$, where the first oscillator is set at frequency f_1 and the second oscillator is at frequency f_2 . It also requires the first band-pass filter (BPF) to have a lower cutoff frequency equal to f_1 and the lower cutoff frequency of the second BPF to be frequency $f_1 + f_2$.



Demonstrate the two-step SSB system's operation by showing the line spectra at points A, B, C and D along the component chain.

It is suggested to use tone modulation to show this; assume $m(t)$ to be at frequency ω_m .

Vestigial Sideband (VSB)

A vestigial filter $H_i(\omega)$ has a transfer function as shown in Figure 4. The carrier frequency is $f_c = 11$ kHz and the baseband signal bandwidth is 5 kHz. **Find** the corresponding transfer function of the equalizer filter $H_o(\omega)$.

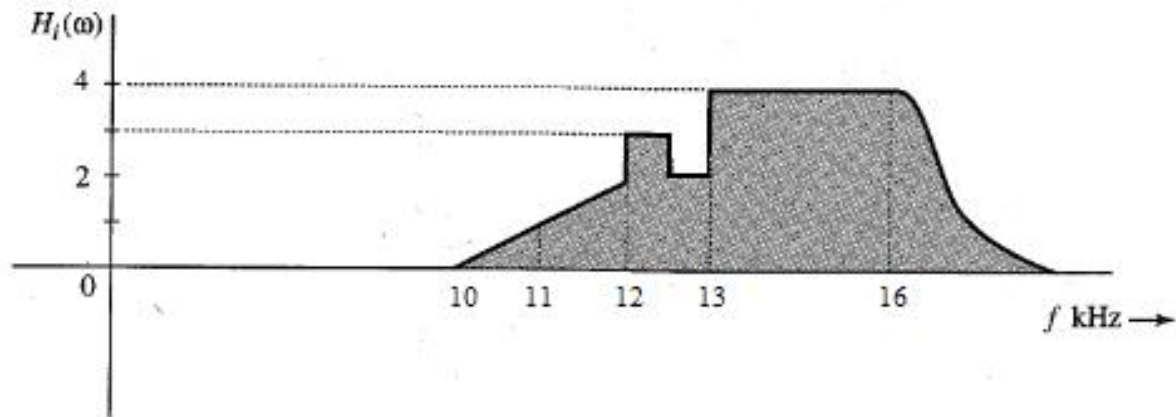


Figure 4. Transfer function of a Vestigial filter